

FELIPE RODRIGUES PEIXOTO DA SILVA

Computer Engineer · Full-Stack Developer

São Paulo, Brazil · frps2003@gmail.com · github.com/Comunalti

SUMMARY

Computer Engineer (Instituto Mauá de Tecnologia, 2025) with 3+ years of industry experience building distributed web platforms, AI pipelines and internal tools end to end — from deployment to pixels. Specialist in data-driven systems: engines and platforms that non-technical people can configure and extend without touching source code. Solid game development background (Unity, shaders, procedural generation) applied to high-quality software engineering.

PROFESSIONAL EXPERIENCE

Full-Stack Developer · Intelicity Dec 2024 — present

- Co-rewrote the company's core AI image-processing pipeline as a distributed system: containers, orchestration and RabbitMQ messaging.
- Rebuilt the image classification platform with Flutter (client), FastAPI (backend) and MongoDB: parallel image downloads, a single flexible classification mode and 100% GUI-based management.
- Single-handedly built, end to end (deploy, Docker, frontend, backend, database), a platform for city governments to register, browse and audit urban roadwork: users, roles, permissions, per-page/per-feature access control and event-triggered scheduled jobs — all dynamically configurable through the UI.

Intern — AI & Smart Cities · Intelicity Mar 2023 — Dec 2024

- Computer vision pipelines in Python: object detection and classification across thousands of images, with georeferenced data in PostgreSQL.
- Created SGC (Sistema Geral de Classificação — General Classification System), an internal image classification tool (Unity/C#) with per-user login, real-time API communication and 4 configurable classification modes — replacing a manual ZIP-based workflow.

HIGHLIGHTED PROJECTS

Data-driven fighting game engine (Thesis) — Unity/C# engine that reads rules from an external XML file and builds characters declaratively, with zero recompilation; serialized state machine + visual editor. Built with a partner Design thesis team (Iralem Studios).

Bilingual full-stack portfolio — React + Node + TypeScript on Docker Compose, PT/EN content served via API and an interactive WebGL/GLSL background shader.

EDUCATION

B.Sc. in Computer Engineering · Instituto Mauá de Tecnologia 2021 — 2025

Mechatronics Technician (integrated with high school) · ETEC Júlio de Mesquita 2018 — 2020

ACADEMIC ACTIVITIES

- Game Development teaching assistant (2023–2024): taught Unity, SOLID, DRY and design patterns to freshmen; lectured on shaders (HLSL, CG/ShaderLab, GLSL) and GPGPU with compute shaders; ran procedural generation workshops (wave function collapse, perlin noise).
- Nawat Games — game development student organization: member (2022), president (2024); Unity/C# games published on itch.io, leading multidisciplinary student teams.

HARD SKILLS

Languages	TypeScript/JavaScript, Python, C#, Dart, SQL, HLSL/GLSL
Backend	Node.js, Express, FastAPI, REST APIs, RabbitMQ, WebSockets
Frontend	React, Vite, Flutter, HTML/CSS, WebGL
Data	PostgreSQL (incl. georeferenced data), MongoDB
DevOps	Docker, Docker Compose, container orchestration, CI/CD, Linux, Git
AI & Vision	Computer vision, object detection/classification, data annotation
Games & Graphics	Unity, data-driven engines, compute shaders, procedural generation (WFC, Perlin)
Architecture	Distributed systems, messaging, RBAC, design patterns (GoF), SOLID, DRY

SOFT SKILLS

Leadership (student organization president)	Teaching and mentoring (2 years as a scholarship TA)	
End-to-end ownership (entire platform built solo)	Cross-disciplinary collaboration (engineering + design)	
Pragmatic problem solving	Continuous, self-driven learning	Clear technical communication
Product thinking for non-technical users		

LANGUAGES

Portuguese — native English — professional (technical reading, writing and documentation)